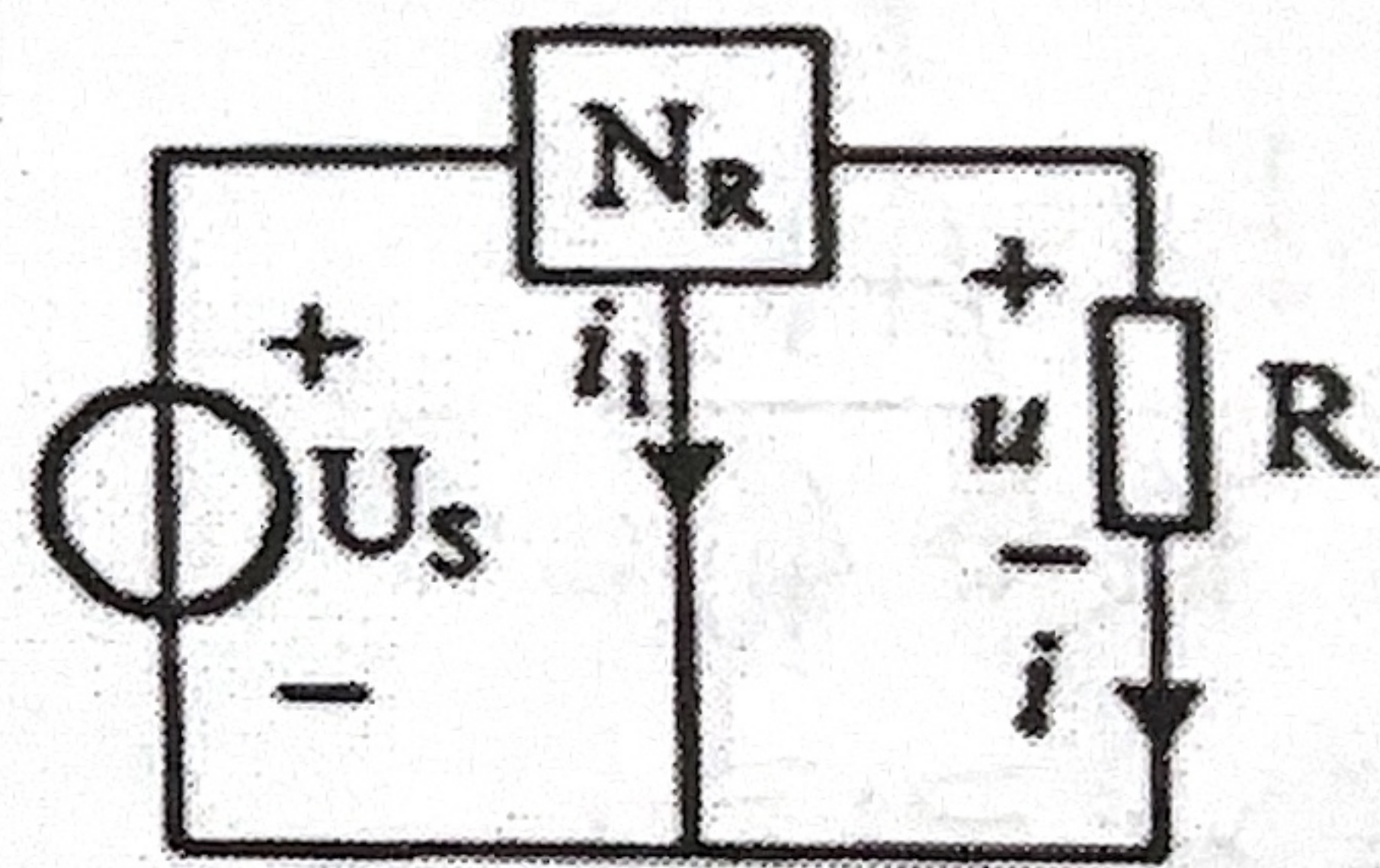


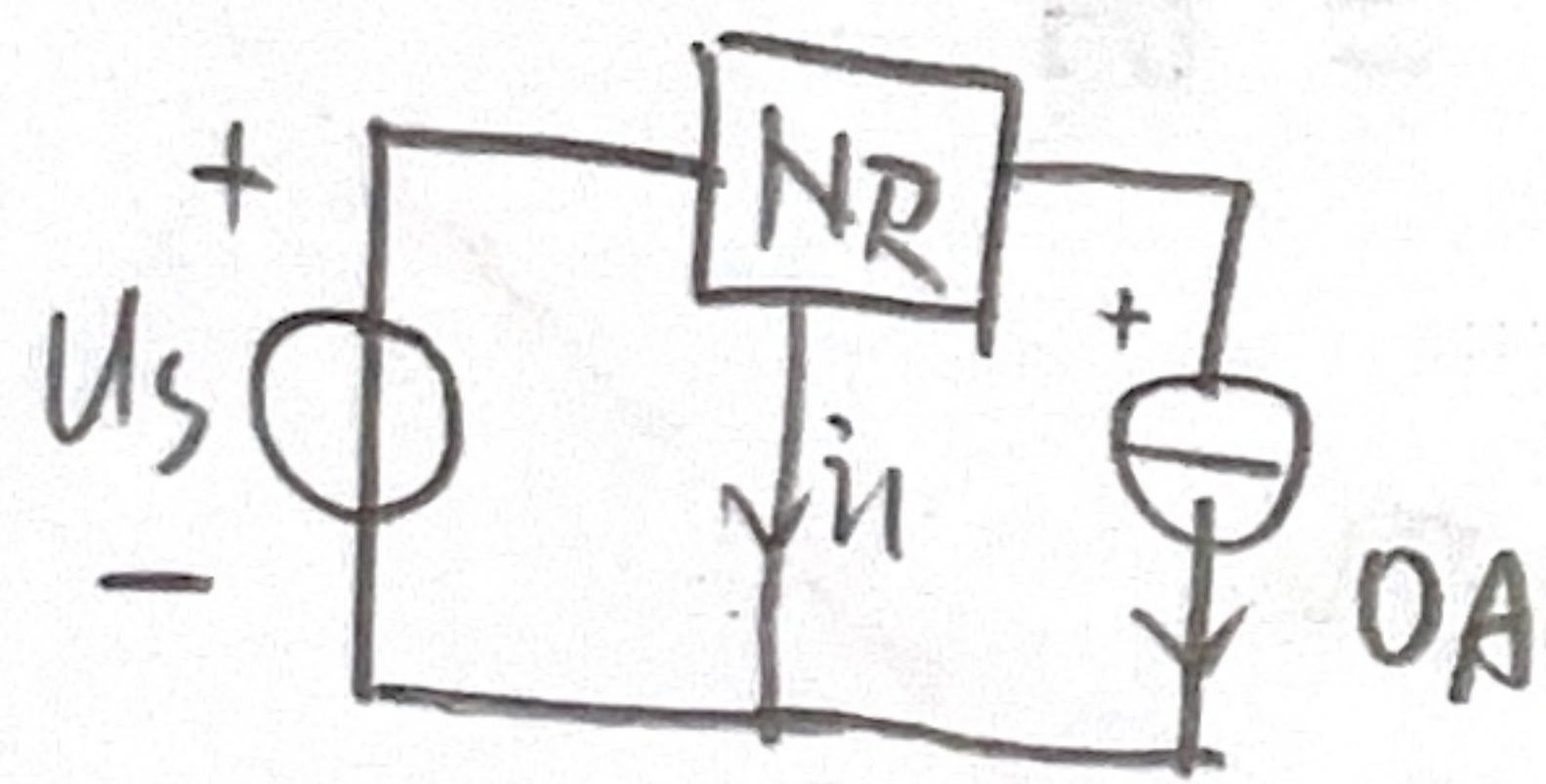
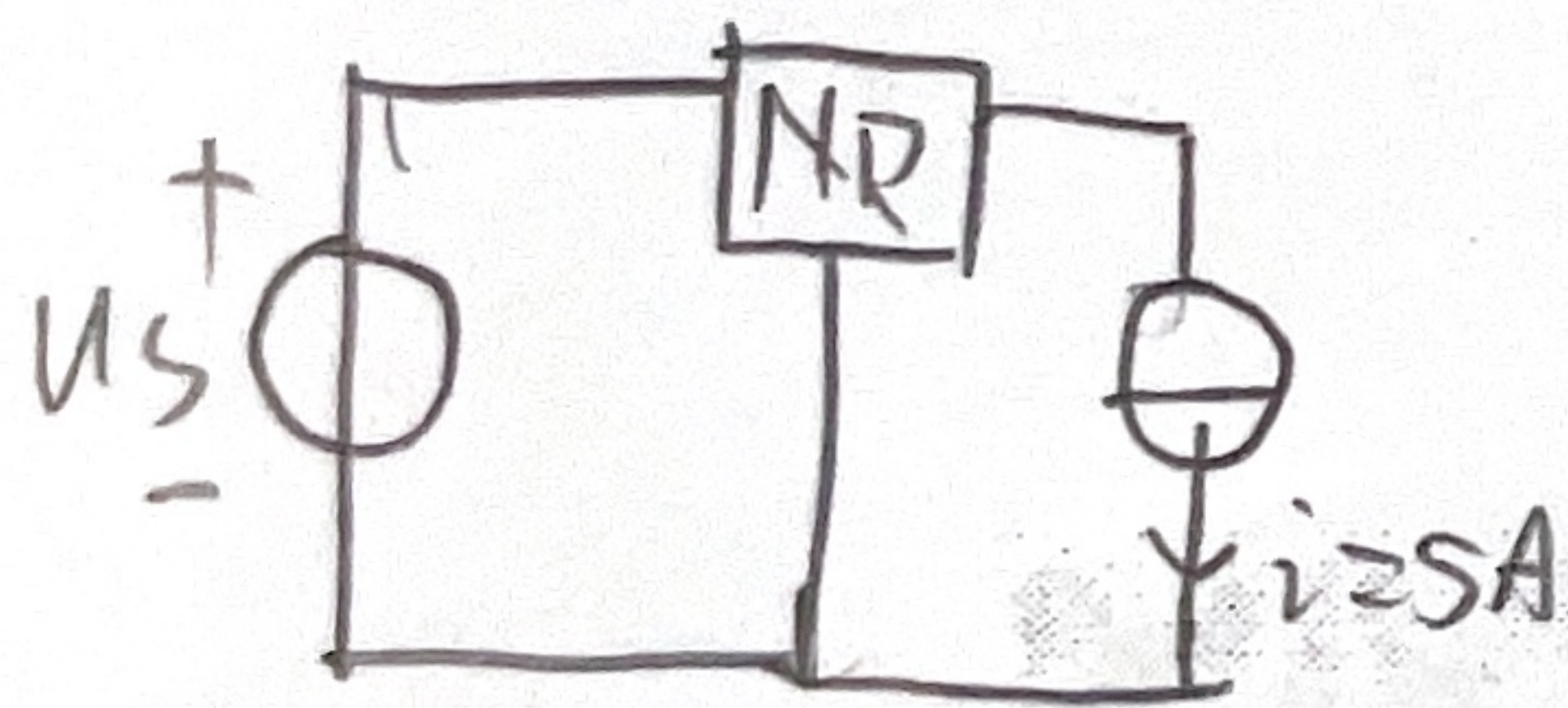
(10分) 11、题 11 图所示电路, N_R 是线性无源电阻网络,
当 $R=0$ 时, $i=5A$, $i_1=4A$; 当 R 趋于 ∞ 时, $u=10V$, $i_1=1A$.

求 $R=R_1$ 时, $i=3A$, i_1 等于多少?



题11图

解: 由替代定理



当 $i=3A$

$$i_1 = \frac{9}{5}A + 1 = \frac{14}{5}A$$

$$i_1 = k_1 i + k_2$$

$$\therefore k_1 = \frac{3}{5}$$

$$4 = 5k_1 + k_2 \quad \text{--- ①}$$

$$\therefore i_1 = \frac{3}{5}i + 1$$